

MATH 2102 Linear Algebra II

Chapter 1 Vector Spaces

Department of Mathematics, HKU

Chapter 1: Vector Spaces

- A The field of scalars
- B Vector spaces and subspaces
- C Basis and dimension
- D New vector spaces from old (I)
- E New vector spaces from old (II)

A1. Vectors vs. scalars

We know that one of the most important objects in linear algebra is a **vector space**. In MATH2101, we focussed on **subspaces of $\mathbb{R}^n$** and we call them **vectors**, in contrast to **scalars** which are usually real numbers (sometimes complex numbers).

In this course, we will adopt a more abstract and general approach to studying linear algebra and vector spaces. Later we will study vector spaces whose elements are not necessarily vectors in $\mathbb{R}^n$. For now, we will focus on the scalars, which are not necessarily real/complex numbers. In general, we can take any **field** and call its elements ‘scalars’. Hence we first need to know what a field is.

A2. Fields

Intuitively, a field is a set on which we can perform addition, subtraction, multiplication, as well as division (except by zero), just like on real numbers.

Formally, a ~~field~~ **field** F is a set equipped with two binary operations, ^①addition (usually denoted by $+$) and ^②multiplication (denoted by $\cdot$ or simply omitted) which satisfy the 'usual properties of arithmetic', including the following:

- (1) There is an **additive identity** 0 , which satisfies $x + 0 = 0 + x = \underline{x}$ for all $x \in F$.
- (2) There is a **multiplicative identity** 1 , which satisfies $x \cdot 1 = 1 \cdot x = \underline{x}$ for all $x \in F$.

If necessary, we may write 0_F and 1_F to refer specifically to the additive identity and multiplicative identity of F , to avoid confusion with the real numbers 0 and 1 (which play the same role on the field of real numbers). It is also assumed that $0 \neq 1$.

A2. Fields

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Formally, a **field** F is a set equipped with two binary operations, addition (usually denoted by $+$) and multiplication (denoted by $\cdot$ or simply omitted) which satisfy the 'usual properties of arithmetic', including the following:

- (3) Addition and multiplication are **commutative**, i.e. $x+y = y+x$ and $x \cdot y = y \cdot x$ for all $x, y \in F$.
- (4) Addition and multiplication are **associative**, i.e. $(x+y)+z = x+(y+z)$ and $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in F$.

A2. Fields

Intuitively, a field is a set on which we can perform addition, subtraction, multiplication, as well as division (except by zero), just like on real numbers.

Formally, a **field** F is a set equipped with two binary operations, addition (usually denoted by $+$) and multiplication (denoted by $\cdot$ or simply omitted) which satisfy the 'usual properties of arithmetic', including the following:

- (5) Every element has an **additive inverse**, i.e. for any $x \in F$, there exists $-x \in F$ such that $x + (-x) = 0_F$.
- (6) Every non-zero element has a **multiplicative inverse**, i.e. for any $x \in F \setminus \{0\}$, there exists $x^{-1} \in F$ such that $x \cdot x^{-1} = 1_F$.
- (7) The **distributive law** holds, i.e. $(x+y) \cdot z = x \cdot z + y \cdot z$ for all $x, y, z \in F$.

A3. Examples and non-examples

(a) $\mathbb{R}$ and $\mathbb{C}$ are fields, with $0_{\mathbb{R}} = 0_{\mathbb{C}} = 0$ and $1_{\mathbb{R}} = 1_{\mathbb{C}} = 1$.

In this course, we almost always assume the field of scalars is $\mathbb{R}$ or $\mathbb{C}$, which we denote by $\mathbb{F}$. Whether $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$ either does not matter, or should be made clear from the context.

A3. Examples and non-examples

(b) $\mathbb{Q}$ is also a field. ✓

How about $\mathbb{Z}$? ✗

$\exists x \in \mathbb{Z}$, no multiplicative inverse

A3. Examples and non-examples

- (c) The smallest field is $\mathbb{F}_2 = \{0, 1\}$. What are its additive and multiplicative structures? ✓

More generally, if p is a prime number, there exists a field with p elements, denoted by $\mathbb{F}_p = \{0, 1, \dots, p-1\}$, with addition and multiplication taken modulo p .

only for $p \nmid$ prime

$$a \times =$$

quiz: (1) ✓ (2) $\mathbb{Z}_1 \cdot \mathbb{Z}_2$ ✓ (4) $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} c & 0 \\ 0 & d \end{bmatrix} = \begin{bmatrix} ac & 0 \\ 0 & bd \end{bmatrix}$ ✓ (8) $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \begin{bmatrix} d & e \\ 0 & f \end{bmatrix} = \begin{bmatrix} ad & ae+bf \\ 0 & cf \end{bmatrix}$ ✓

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B1. Examples for motivation

In MATH2101, we learnt that **addition** and **scalar multiplication** are the two key operations on vector spaces. On the other hand, in Part A we mentioned that we can perform **addition** and **multiplication** on a field. Can we put these together?

For a field F and positive integer n , we use F^n to denote the set of ordered n -tuples of F , e.g. we have the following familiar examples:

- $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$
- $\mathbb{C}^3 = \{(x, y, z) : x, y, z \in \mathbb{C}\}$

We also use F^∞ to denote the set of infinite sequences in F . For instance, $\mathbb{F}_2^\infty$ denotes the set of infinite binary strings.

B1. Examples for motivation

With the notations on the previous page, we can naturally define addition and scalar multiplication on F^n (also F^∞ in a similar manner) **componentwise**. If $\lambda \in F$, $\mathbf{x} = (x_1, x_2, \dots, x_n) \in F^n$ and $\mathbf{y} = (y_1, y_2, \dots, y_n) \in F^n$, then we may define

$$\mathbf{x} + \mathbf{y} = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$$

and

$$\lambda \mathbf{x} = (\lambda x_1, \lambda x_2, \dots, \lambda x_n).$$

Note that $\mathbf{x} + \mathbf{y}$ and $\lambda \mathbf{x}$ are both in F^n . (Why?)

B2. Definition of a vector space

Let V be a set (whose elements are called **vectors**) and F be a field (whose elements are called **scalars**). We say that V is a **vector space over F** (or simply an **F -vector space**) (if we can define the two operations below which satisfy the requirements on the next page:)

- **Addition** of two vectors that gives a vector, i.e. for any $\mathbf{u}, \mathbf{v} \in V$, the sum $\mathbf{u} + \mathbf{v}$ is in V .
- **Scalar multiplication** of a vector by a scalar that gives a vector, i.e. for any $\lambda \in F$ and $\mathbf{v} \in V$, the product $\lambda \mathbf{v}$ is in V .

B2. Definition of a vector space

Requirements for the two operations:

- Addition is commutative and associative.
- There exists an **additive identity**.
- Every element of V has an **additive inverse**, i.e. for all $\mathbf{v} \in V$ there exists $-\mathbf{v} \in V$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.
- For any $\mathbf{v} \in V$, we have $1\mathbf{v} = \mathbf{v}$.
- The **distributive law** holds, i.e. for any $\alpha, \beta \in F$ and $\mathbf{u}, \mathbf{v} \in V$, we have $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$ and $(\alpha + \beta)\mathbf{u} = \alpha\mathbf{u} + \beta\mathbf{u}$.

B3. Examples and non-examples

- (a) In view of the discussions in B1, we know that F^n and F^∞ are vector spaces over F .

In particular, $\mathbb{R}^n$ is a vector space over $\mathbb{R}$, a familiar example from MATH2101.

Similarly, $\mathbb{C}^n$ is a vector space over $\mathbb{C}$. Is $\mathbb{C}^n$ is a vector space over $\mathbb{R}$? 

But $\mathbb{R}^n$ is not a v- over $\mathbb{C}$

B3. Examples and non-examples

- (b) $\mathbb{F}_2^\infty$, the set of infinite binary strings, is a vector space over $\mathbb{F}_2$. How are addition and scalar multiplication defined?

B3. Examples and non-examples

(c) Let $F[x]$ denote the set of all polynomials in x with coefficients in F . Then $\mathbb{R}[x]$ is a vector space over $\mathbb{R}$.

$P_n(\mathbb{R})$, the set of polynomials with coefficients in $\mathbb{R}$ whose degree is at most n , is also a vector space over $\mathbb{R}$.

How about the set of polynomials with coefficients in $\mathbb{R}$ whose degree is equal to n ?

B3. Examples and non-examples

(d) Let $\mathbb{F}^{m,n}$ denote the set of $m \times n$ matrices with entries in $\mathbb{F}$. Then $\mathbb{F}^{m,n}$ is a vector space over $\mathbb{F}$.

(Some alternative notations are $\mathbb{F}^{m \times n}$ and $M_{m \times n}(\mathbb{F})$.)

B3. Examples and non-examples

- (e) Let $\mathcal{C}(I)$ denote the set of continuous functions with domain I . Then $\mathcal{C}(\mathbb{R})$ is a vector space over $\mathbb{R}$.

B4. Subspaces

Let V be a vector space over F . A subset W of V which is also a vector space over F is called a **subspace** of V . This is sometimes denoted as $W \leq V$.

If $W \subseteq V$, to check whether W is a subspace of V it suffices to check three things:

- 0 $\in W$.
- For all $u, v \in W$, we have $u + v \in W$.
- For all $\lambda \in F$ and $u \in W$, we have $\lambda u \in W$.

B4. Subspaces

In each of the following, is W a subspace of V ?

- (a) ~~(0)~~/x $V = P_n(\mathbb{C})$, $W =$ set of polynomials in V with constant term zero
- (b) ✓/~~0~~ $V = \mathcal{C}(\mathbb{R})$, $W = \{f(x) \in V : |f(x)| \leq 1 \text{ for all } x \in \mathbb{R}\}$
- (c) ✓/~~0~~ $V = \mathbb{F}_2^\infty$, $W =$ set of strings in V with finitely many 1's

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C1. Span and linear independence

In MATH2101 we learnt about the concepts of span and linear independence in $\mathbb{R}^n$. These are exactly the same in general vector spaces, except that the ‘vectors’ are not necessarily from $\mathbb{R}^n$ and the ‘scalars’ are not necessarily from $\mathbb{R}$.

More precisely, let V be a vector space over F , $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a subset of V and $c_1, c_2, \dots, c_n \in F$.

- The **span** of S (denoted **Span S**) is defined to be the set of all **linear combinations** of the vectors in S , i.e. vectors of the form $\sum c_i \mathbf{v}_i$ where $c_1, c_2, \dots, c_n \in F$.
- We say that S is **linearly dependent** if there exist $c_1, c_2, \dots, c_n \in F$, not all zeros, such that $\sum c_i \mathbf{v}_i = \mathbf{0}$. Otherwise S is **linearly independent**.

C2. Basis

The notion for **basis** is the same in a general vector space as in $\mathbb{R}^n$, namely, a basis for V is simply a linearly independent subset S of V whose span is V . (By convention, $\emptyset$ is regarded to be a basis for $\{\mathbf{0}\}$.)

However, unlike subspaces of $\mathbb{R}^n$, not every vector space has a finite basis. In particular, we say that a vector space V is

- **finite-dimensional** if it has a finite basis; and
- **infinite-dimensional** otherwise.

The study of infinite-dimensional vector spaces would involve the span of an infinite set, which shall not be the focus of this course. We shall however look at a couple of examples of infinite-dimensional vector spaces.

C2. Basis

Show that the following vector spaces are infinite-dimensional.

- (a) $\mathbb{R}[x]$ over $\mathbb{R}$
- (b) $\mathbb{F}_2^\infty$ over $\mathbb{F}_2$

Proof:

In each case assume a finite basis exists, and we shall derive a contradiction.

- (a) Suppose $\{p_1(x), p_2(x), \dots, p_n(x)\}$ is a basis.

$$\deg\left(\sum p_i(x) c_i\right) \leq \max_{p_i(x)} \deg p_i(x)$$

- (b) Suppose $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis.

C3. Dimension

From now on,

All vector spaces are assumed to be finite-dimensional, unless otherwise specified or when an explicit infinite-dimensional example is given.

As in the case of subspaces of $\mathbb{R}^n$, we have the following result.

Let V be a vector space. Then any two bases of V have the same size, which is known as the **dimension** of V , denoted by **$\dim V$** .

If necessary we write $\dim_F V$ to emphasise that the field of scalars is F .

C3. Dimension

The proof that any two bases of V have the same size have two main steps:

- If S is a linear dependent subset of V , then there exists $\mathbf{v} \in S$ such that $\mathbf{v} \in \text{Span}(S - \mathbf{v}) = \text{Span } S$.
- If S is a linear independent subset of V and T is a spanning set for V , then $|S| \leq |T|$.

Let $V = \{v_1, \dots, v_n\}$ $W = \{w_1, \dots, w_m\}$ $\rightarrow m \geq n$

so $\{w_1, \dots, w_m\}$ is spanning set for S

i.e. $\{v_1, w_1, \dots, w_m\}$ is $\sim$
 $\rightarrow$ linearly dependent

we discard w_1

$\{v_1, w_2, \dots, w_m\}$ is spanning set for S .

...

$\{v_1, v_2, \dots, v_n, \underbrace{w_{n+1}, \dots, w_m}_{\text{discarded}}\}$ is $\sim$

C4. Examples

Find the dimensions of the following vector spaces.

(a) $\mathbb{C}^4$ over $\mathbb{C}$ 4

(b) $\mathbb{C}^4$ over $\mathbb{R}$ 8

(c) $P_n(\mathbb{R})$ over $\mathbb{R}$ $n+1$

(d) $\mathbb{C}^{2 \times 3}$ over $\mathbb{R}$ 12

C5. Further remarks

We have learnt the following results if V is a subspace of $\mathbb{R}^n$ with dimension k and $S \subseteq V$. They still hold if V is a general vector space of dimension k . The proof is essentially the same.

- Suppose $|S| < k$. Then S cannot span V . If S is linearly independent, it can be extended to a basis of V .
- Suppose $|S| > k$. Then S cannot be linearly independent. If S spans V , it can be reduced to a basis of V .
- Suppose $|S| = k$. Then S is linearly independent if and only if it spans V .

Chapter 1: Vector Spaces

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D1. Intersection and union

In MATH2101 we learnt that:

- the intersection of two subspaces of $\mathbb{R}^n$ is again a subspace;
- the union of two subspaces of $\mathbb{R}^n$ is not necessarily a subspace.

These are also true for general vector spaces. More precisely, given a vector space V and two subspaces W_1 and W_2 of V ,

- 
- $W_1 \cap W_2$ is a subspace of V ;
 - $W_1 \cup W_2$ not necessarily a subspace.

D2. Sum of vector spaces

Since $W_1 \cup W_2$ may not be a vector space in general, we would like to find a vector space containing both W_1 and W_2 .

Let W_1 and W_2 be subspaces of a vector space V . The **sum** of W_1 and W_2 , denoted by $W_1 + W_2$ and defined by

$$W_1 + W_2 = \{\mathbf{w}_1 + \mathbf{w}_2 : \mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2\},$$

is also a subspace of V .

D2. Sum of vector spaces

Consider the following subspaces of $\mathbb{F}^4$. Can you find $U_1 + U_2$ and $V_1 + V_2$?

- $U_1 = \{(x, x, 0, 0) : x \in \mathbb{F}\}$
- $U_2 = \{(0, 0, 0, x) : x \in \mathbb{F}\}$
- $V_1 = \{(x, x, y, y) : x, y \in \mathbb{F}\}$
- $V_2 = \{(x, x, x, y) : x, y \in \mathbb{F}\}$

$$U_1 + U_2 = \{(x, x, 0, y) : x, y \in \mathbb{F}\}$$

$$V_1 + V_2 = \{(x, x, y, z) : x, y, z \in \mathbb{F}\}$$

D2. Sum of vector spaces

The dimension of the sum satisfies the following formula:

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$$

Proof:

Let $\dim(W_1 \cap W_2) = d$ and $\mathcal{B}$ be a basis of $W_1 \cap W_2$. Extend $\mathcal{B}$ to a basis $\mathcal{B}_1$ of W_1 and a basis $\mathcal{B}_2$ of W_2 , with $|\mathcal{B}_1| = d + e_1$ and $|\mathcal{B}_2| = d + e_2$. We want to show that $\dim(W_1 + W_2) = \underline{d + e_1 + e_2}$. A natural choice of basis would be

D3. Direct sum

If $W_1 \cap W_2 = \{\mathbf{0}\}$, then the proof on the previous page suggests that we can simply put together a basis of W_1 and a basis of W_2 to form a basis of $W_1 + W_2$. It looks like a 'disjoint union' of W_1 and W_2 .

We say that $W_1 + W_2$ is a **direct sum**, denoted $W_1 \oplus W_2$, if $W_1 \cap W_2 = \{\mathbf{0}\}$. Can you find examples and non-examples?

D3. Direct sum

The following is an equivalent condition for a sum of subspaces to be direct. Many books also use this as the definition of a direct sum.

Let W_1 and W_2 be subspaces of a vector space V . Then $W_1 + W_2$ is a direct sum if and only if every vector $\mathbf{w} \in W_1 + W_2$ can be expressed in the form $\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2$, where $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$, in a **unique** way.

$$V = W_1 \oplus W_2$$

$$W_1 \cap W_2 = \{0\}$$

D4. Sums of more than two subspaces

Naturally, we can extend the notion of the sum and direct sum of two subspaces to three or more subspaces. More precisely, let $W_1, W_2, \dots, W_k$ be subspaces of a vector space V . Then we define

$$W_1 + W_2 + \cdots + W_k = \{\mathbf{w}_1 + \mathbf{w}_2 + \cdots + \mathbf{w}_k : \mathbf{w}_i \in W_i \text{ for } 1 \leq i \leq k\}.$$

D4. Sums of more than two subspaces

The following statements regarding $W_1 + W_2 + \cdots + W_k$ are equivalent, and if any one of them holds then the sum is a direct sum, denoted by $W_1 \oplus W_2 \oplus \cdots \oplus W_k$

~~(1) $W_i \cap W_j = \{0\}$ for any $1 \leq i \neq j \leq k$.~~

$$\forall i, U_i \cap \left(\sum_{j \neq i} U_j \right) = \{0\}$$

(2) $\dim(W_1 + W_2 + \cdots + W_k) = \dim W_1 + \dim W_2 + \cdots + \dim W_k$.

(3) Any vector $\mathbf{w}$ can be expressed uniquely in the form $\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2 + \cdots + \mathbf{w}_k$, where $\mathbf{w}_i \in W_i$ for $1 \leq i \leq k$.

(4) The uniqueness condition in (3) holds for $\mathbf{w} = \mathbf{0}$.

(5) B_i is basis of W_i . $\bigcup_{i=1}^k B_i$ is linear independent.

⊂ The basis of $S = W_1 \oplus W_2 \cdots \oplus W_k$

$$\begin{cases} \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} \\ \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \\ \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} \end{cases}$$

Chapter 1: Vector Spaces

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E1. Product of vector spaces

If V and W are **sets**, we can form the **Cartesian product**

$$V \times W = \{(\mathbf{v}, \mathbf{w}) : \mathbf{v} \in V, \mathbf{w} \in W\}.$$

If V and W are **vector spaces** over the same field F , then the Cartesian product $V \times W$ will also be a vector space over F , since we can define addition and scalar multiplication **componentwise**:

$$(\mathbf{v}_1, \mathbf{w}_1) + (\mathbf{v}_2, \mathbf{w}_2) = (\mathbf{v}_1 + \mathbf{v}_2, \mathbf{w}_1 + \mathbf{w}_2)$$

$$\lambda \cdot (\mathbf{v}, \mathbf{w}) = (\lambda \mathbf{v}, \lambda \mathbf{w})$$

This also naturally extends to the product of three or more vector spaces.

E1. Product of vector spaces

Unlike the sum, where we have to add two subspaces under the same vector space, we can take the product of two 'very different' vector spaces, as long as they are vector spaces over the same field F .

For example, let $V = \mathbb{R}^4$ and $W = P_3(\mathbb{R})$, both considered as vector spaces over $\mathbb{R}$. Then we can define $V \times W$.

E1. Product of vector spaces

We have $\dim(V \times W) = \dim V$ + $\dim W$.

Proof:

Let $\dim V = d_1$ and $\dim W = d_2$, with $\mathcal{B}_1$ and $\mathcal{B}_2$ being bases of V and W respectively. Then

$$\text{Let } \mathcal{B}_1 = \{a_1, \dots, a_{d_1}\}$$

$$\mathcal{B}_2 = \{b_1, \dots, b_{d_2}\}$$

then the basis of $V \times W$ is : $\{ (a_1, 0), \dots, (a_{d_1}, 0), \\ (0, b_1), \dots, (0, b_{d_2}) \}$.

E2. Motivation for quotient spaces

What is the unit digit of $23456 + 34567$?

This is like, we can partition all non-negative integers into 10 subsets ‘modulo 10’. We can take a representative for each set, say, using 0 to 9 to represent the sets. Then we can ‘add two sets’, e.g. $[6] + [7] = [3]$.

E2. Motivation for quotient spaces

We can likewise partition $\mathbb{R}^2$ into (infinitely many) straight lines (subsets of $\mathbb{R}^2$) with slope 2. We can take $(0, c)$ to represent each line, where c is the y -intercept. This time we can ‘add two lines’ as well as ‘multiply a line by a scalar’ to get a line (with slope 2), e.g.

- adding the line with y -intercept 3 to the line with y -intercept 4 gives the line with y -intercept 7;
- multiplying the line with y -intercept 3 by the scalar -1.5 gives the line with y -intercept -4.5 .

Such addition and scalar multiplication of lines seem rather meaningless at first glance. However, it makes good sense if we pick a point (vector) on the lines and add/scalar multiply the vectors.

E3. Quotient of vector spaces

Let V be a vector space and W be a subspace of V . For any $\mathbf{v} \in V$, define

$$\mathbf{v} + W = \{\mathbf{v} + \mathbf{w} : \mathbf{w} \in W\}.$$

This is known as a **translate** of W .

For instance, if $V = \mathbb{R}^2$ and $W = \{(x, 2x) : x \in \mathbb{R}\}$, what are $(1, 1) + W$, $(2, 2) + W$ and $(2, 3) + W$?

E3. Quotient of vector spaces

As we noted, there are different ways of representing a translate.

We have $\mathbf{v}_1 + W = \mathbf{v}_2 + W$ if and only if $\mathbf{v}_1 - \mathbf{v}_2 \in W$.

$$\textcircled{1} \exists w_1, w_2 \in W \quad v_1 + w_1 = v_2 + w_2$$

$$v_1 - v_2 = w_2 - w_1 \in W$$

E3. Quotient of vector spaces

We can add two translates or multiply one by a scalar using the following definitions:

$$\begin{aligned}
 & \text{Handwritten: } v_1 + w_1 + v_2 + w_2 = (v_1 + v_2) + (w_1 + w_2) \\
 & (v_1 + W) + (v_2 + W) = \underline{(v_1 + v_2) + W} \\
 & \lambda \cdot (v + W) = \underline{(\lambda v) + W}
 \end{aligned}$$

What is a potential problem with such definition?

E3. Quotient of vector spaces

Let V be a vector space over F and W be a subspace of V . Denote the set of all translates of W by V/W , i.e.

$$V/W = \{\mathbf{v} + W : \mathbf{v} \in V\}.$$

Then V/W is a vector space over F , known as a **quotient space**. We shall investigate the dimension of V/W in the next chapter.